

MTH 732 Topology - Homework 10

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Assignment 1. Show that the degree of the antipodal map $A : S^n \rightarrow S^n$ given by $A(x) = -x$ is $\deg(A) = (-1)^{n+1}$.

Proof. First notice that if we only “flip” one of the coordinates with a map f , then f has degree -1. Composing maps $S^n \rightarrow S^n$ causes the degrees to multiply, so if we consider A to be the composition of $n + 1$ single-coordinate flips (letting S^n sit inside R^{n+1}), then we get that $\deg(A) = (-1)^{n+1}$ as desired. $\square$

Assignment 2. Describe a CW complex structure on the 3-torus $T^3 = S^1 \times S^1 \times S^1$ and compute its homology.

Proof. Using the product CW structure, we can see that since $S^1 \times S^1$ has one 0-cell, two 1-cells, and one 2-cell and S^1 has one 0-cell and one 1-cell, the product will give one 0-cell, three 1-cells, three 2-cells, and one 3-cell. The 1-cells are attached by the only possible map, namely the constant map, so ∂_1 is the zero map. The 2-cells are attached similar to on the torus, where each 1-cell is traversed twice, once backward and once forward, so ∂_2 is also the zero map. Finally, the 3-cell connects all of the 2-cells, and once again in a way that traverses each one once in each orientation, so ∂_3 is also zero. Thus we can read off the homology straight from the chain groups and we get

$$H_n(T^3) = \begin{cases} \mathbb{Z} & n = 0, 3 \\ \mathbb{Z}^3 & n = 1, 2 \\ 0 & \text{otherwise.} \end{cases}$$

$\square$

Assignment 3. For $n < m$, find the homology groups of the quotient space $\mathbb{RP}^m/\mathbb{RP}^n$, where $\mathbb{RP}^n$ sits inside $\mathbb{RP}^m$ as its n -skeleton.

Proof. First, note that $\mathbb{RP}^m$ has one cell in each dimension from 0 through m , so $\mathbb{RP}^m/\mathbb{RP}^n$ has one cell in each dimension from $n + 1$ through m , and one 0-cell (which arises from the modded-out $\mathbb{RP}^n$). So the cellular chain complex looks like this:

$$\dots \xrightarrow{\partial_{m+2}} 0 \xrightarrow{\partial_{m+1}} \mathbb{Z} \xrightarrow{\partial_m} \dots \xrightarrow{\partial_{n+2}} \mathbb{Z} \xrightarrow{\partial_{n+1}} 0 \xrightarrow{\partial_n} \dots \xrightarrow{\partial_2} 0 \xrightarrow{\partial_1} \mathbb{Z} \xrightarrow{\partial_0} 0 ,$$

where the boundary maps are all 0 since the attaching maps are all constant maps to the 0-cell. Given this, we can read off

$$H_k(\mathbb{RP}^m/\mathbb{RP}^n) = \begin{cases} \mathbb{Z} & k = 1 \\ \mathbb{Z} & n+1 \leq k \leq m \\ 0 & \text{otherwise.} \end{cases}$$

□

Assignment 4. In class, we saw that the real Grassmannian $\text{Gr}(2, 4)$ has a CW complex structure coming from all possible reduced row echelon forms of 2×4 matrices. In particular, there are exactly two 2-cells:

$$e_1^2 = \begin{pmatrix} 0 & 1 & 0 & x \\ 0 & 0 & 1 & y \end{pmatrix} \quad \text{and} \quad e_2^2 = \begin{pmatrix} 1 & x & y & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

where $(x, y) \in \mathbb{R}^2$. Show that the attaching maps $\varphi_i^2 : \partial D_i^2 \rightarrow X^1$, $i = 1, 2$, of these cells onto the 1-skeleton of $\text{Gr}(2, 4)$ (which is a circle) have degree 2.

Proof. We are given the two 2-cells e_1^2 and e_2^2 , and we also have the 1-cell and 0-cell

$$e^1 = \begin{pmatrix} 0 & 1 & x & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad e^0 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Since e^0 , e^1 , and e_1^2 each have zeroes in the first column, they form a copy of $\text{Gr}(2, 3)$, which is isomorphic to $\text{Gr}(1, 3)$, which is $\mathbb{RP}^2$. We already know that the boundary map of the 2-cell in $\mathbb{RP}^2$ has degree 2, so we're done φ_1^2 .

Now we notice that the second row of e_2^2 is constant, so that any plane represented by e_2^2 contains the vector $(0, 0, 0, 1)$. Thus taking e_2^2 together with e^1 and e^0 , we have again the set of planes in a 3-dimensional subspace, and by the same argument above φ_2^2 has degree 2. □

Assignment 5. Show that any finite, path-connected CW complex X is homotopy equivalent to a CW complex Y having exactly **one** 0-cell.

Proof. Since X is path-connected, there cannot be any isolated 0-cells, since that would produce multiple path components. Thus all the 0-cells have to be connected by some path of 1-cells in the 1-skeleton. We can choose, then, some spanning tree of the 1-skeleton and contract it to get a homotopy equivalent CW complex which has exactly one 0-cell. Note that the homotopy equivalence comes from the fact that the tree itself is contractible since it has no loops, so contracting it gives a homotopy equivalent complex. □

Assignment 6. If $p : \widetilde{X} \rightarrow X$ is a k -fold covering map (every preimage $p^{-1}(x)$ has k elements) and X is a finite CW complex, show that $\widetilde{X}$ has a natural CW complex structure coming from the lifts of the cells of X (more precisely, lifting the characteristic maps ϕ_α^n restricted to $\text{int}(D_\alpha^n)$). Conclude that $\chi(\widetilde{X}) = k\chi(X)$.

Proof. For any x in the interior of some n -cell D_α^n , p restricts to a homeomorphism on any component of $p^{-1}(\text{int}(D_\alpha^n))$. Thus there are k n -cells in $\widetilde{X}$ for each n -cell in X . Meanwhile, if x is on the boundary of some n -cell, then it is also on the boundary of some k -cell for $k < n$ (for at least one such k , but possibly more), so that any neighborhood U of x intersects the interior of all such cells it is on the boundary of. Then the attaching maps pull back to attaching maps in $\widetilde{X}$, and we get that, for a choice of preimage component, we have an entire copy of X of which it is a part. Thus, as desired, we get that $\chi(\widetilde{X}) = k\chi(X)$ since the number of cells in each dimension is multiplied by k , and then we factor k out of the alternating sum. $\square$

Assignment 7. Show that tensoring the short exact sequence $0 \rightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \rightarrow \mathbb{Z}_n \rightarrow 0$ with an Abelian group G can destroy exactness on the left (find such a group G).

Proof. Choose $G = \mathbb{Z}_n$. Then the sequence becomes $0 \rightarrow \mathbb{Z}_n \xrightarrow{n} \mathbb{Z}_n \rightarrow \mathbb{Z}_n \otimes \mathbb{Z}_n \rightarrow 0$. Since the multiplication by n map sends 1 to 0, it is not injective, so the sequence is not left exact. $\square$

Assignment 8. Show that $\mathbb{RP}^3$ and $\mathbb{RP}^2 \vee S^3$ are not homotopy equivalent.

Proof. I know this has something to do with the cup product, but I'm not getting there with this one. $\square$